



features ■ technology next

The Future of



Popular culture has always been bombarded with the thoughts of science fiction writers and pop directors. Whether it is mass-hysteria created by the radio transmission of H.G. Wells' *War of the Worlds*, or the drones of fans that religiously subscribe to the worlds of *Star Trek* and *Star Wars*, fiction has always teased and appeased the human mind. Cars that can talk and drive themselves, buildings that fly, robots that help around the house, and androids that are almost human, universal translators, aircraft that do all the flying for us... joining all these seemingly disparate dots is the common thread of Artificial Intelligence. The concept that we, as humans, will one day create entities that will be non-differentiable from us in thought and in action. Will the line between fact and fiction be always a thick bold one? Are we doomed to conjure up things larger than life whilst we struggle to cure the common cold? Let us see whether science today has succeeded in rendering some of our fondest dreams into reality, even as we look further, into the brave new world of the future. ■

AHMED SHAIKH

Sleek, black, fast and talkative. Riding autonomously on all fours with an indestructible body to boot. This was KITT, a Pontiac Trans-Am and the star of the 1982 television series *Knight Rider*. Can we ever hope to own a car that drives itself?

The bad news is that KITT is still years away—as of now, you cannot hope to control your car with a cell phone. The good news is that there are autonomous systems that allow some amount of back seat driving. Typically such systems, such as the Distronic found in the Mercedes S Class of cars, do little but regulate the distance between vehicles, speeding and slowing as and when required. The better news is that something a little more evolutionary, that can cope with the urban traffic, where lane rules and road regulations are easily forgotten, is in the making. This new car contains two microprocessors that figure out where the car is in relation to others and to the boundaries of the road. The vehicle is driven by software AI techniques of shape recognition and comparison. It's based on a database consisting of three-dimensional models of roads and road elements. Thus, facets of road intersections, types, typical road-widths, cars,

trucks and pedestrians are all allocated within the database. Mounting front of the car are cameras that take snapshots of the surrounding environment and send them to the processor nestled within. The processor uses the AI routines to compare the features to the ones within the database. The car will thus ascertain its position and speed with respect to the vehicles, the road and pedestrian. The processor then steers, accelerates, decelerates as needed. The system is not perfect since the decision is based on ideal road widths and road conditions best on slow speeds.

Future cars could share a common consciousness of sorts; a distributed software environment reminiscent of the Borg collective. Akin to the intelligent behaviour of birds, AI routines could render this collective to act in a coordinated manner, keeping distance, maintaining a safe speed, and ensuring that you reach your destination safely and effectively.